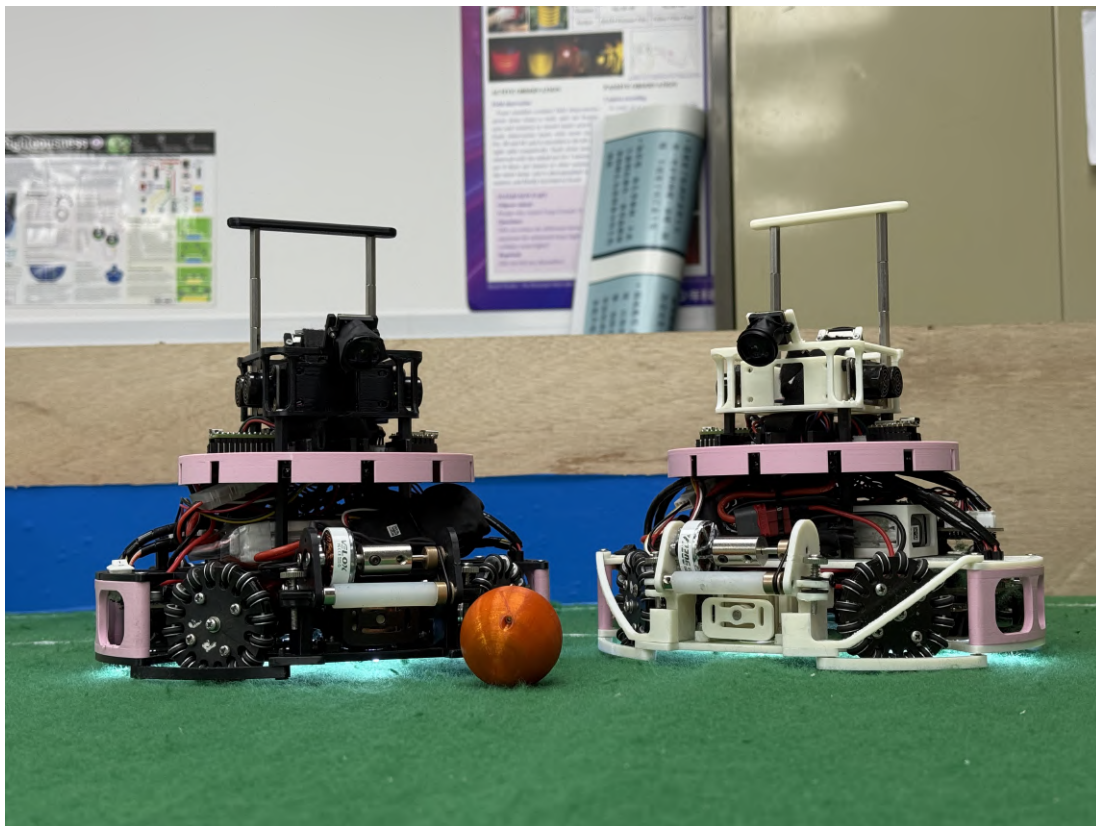


RIGHTEOUSNESS

Technical Design Paper

RoboCupJunior Soccer Infrared 2026



Pui Ching Middle School
Macau, Macao

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ABSTRACT

This Technical Design Paper details the mechanical, electrical, and software engineering innovations developed by Team Righteousness from Pui Ching Middle School Macau for our robot competing in the 2026 RoboCupJunior Soccer infrared competition. To achieve optimal performance within the strict 1.5 kg and 22 cm constraints, our team developed a highly rigid, lightweight Acrylonitrile Styrene Acrylate (ASA) chassis using a cloud-based collaborative Onshape workflow. The electronic architecture features a robust power distribution system and a unique quad-core distributed processing network—utilizing a Teensy 4.1 for core kinematics, a Teensy 4.0 for quad-camera vision aggregation, and dual Seeeduino XIAO ESP32-S3 coprocessors paired with custom hardware low-pass filters for noise-free infrared and boundary line detection. Furthermore, we outline our dynamic state-machine algorithms, highlighting our closed-loop PID holonomic control, advanced active vector containment for defensive boundaries, and adaptive offensive shielding maneuvers.

I. INTRODUCTION

A. Team Background

Team RIGHTEOUSNESS, representing Pui Ching Middle School in Macau, brings together a dedicated group of students with a deep passion for robotics and embedded systems engineering. Building upon previous iterations and rigorous testing, our team has continuously refined our technical capabilities and collaborative workflows to meet the evolving challenges of the RoboCupJunior league.

B. Objectives

For the 2026 RoboCupJunior Soccer Lightweight competition, our primary objective is to engineer a highly agile and deeply integrated robotic platform that pushes the boundaries of distributed processing and autonomous dynamic control. Rather than simply meeting physical constraints, our focus shifted toward maximizing computational efficiency and kinematic fluidity. Our ultimate goal is to eliminate the processing bottlenecks typically associated with high-density sensor arrays and 360-degree environmental awareness, creating a robot capable of instantaneous, mathematically precise reactions during chaotic gameplay. By achieving a seamless synergy between our custom mechanical architecture and our quad-core electronic network, we aim to deliver flawless holonomic maneuverability and highly reactive strategic execution on the field.

MENTOR AND MEMBER INFORMATION**[Ellen Leong]***Team Mentor*

Structuring complex problems, implementing systematic debugging frameworks, and leveraging AI tools for autonomous troubleshooting.

[Yogo Hoi]*Electronics & Defensive Strategy*

Designed custom PCBs, managed electronic architecture, and sourced components. Developed the defender's reactive state machine and vector-based boundary containment algorithms.

[Terris Cheang]*Kinematics & Offensive Algorithm*

Engineered core holonomic kinematics with PID control and IMU yaw compensation. Architected the Attacker's dynamic blended tracking algorithm for optimal, collision-free ball interception.

[Anson Kun]*SrMechanical Design & Hardware Integration*

Designed the lightweight ASA chassis and structural parts using Onshape. Executed precision PCB soldering, system assembly, and co-developed defensive tactical logic.

II. MECHANICAL AND STRUCTURAL HARDWARE

A. Cloud-Based CAD Workflow & Chassis Optimization

Our team utilized a multi-tier software approach to design our robot's mechanical parts, beginning with AutoCAD and CorelDRAW for 2D flat-profile planning before transitioning to Onshape as our primary 3D CAD platform. We selected Onshape for its powerful cloud-based architecture, enabling real-time online collaboration that accelerated our workflow. To maintain a low center of gravity and maximize stability, we positioned the motors as close to the chassis' outer perimeter as possible, while strictly adhering to the 2026 Soccer Lightweight 22 cm and 1.5 kg limits.

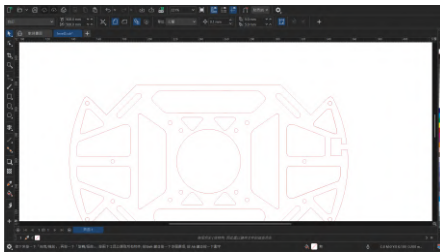


Fig. 1. Initial 2D flat-profile planning of the chassis structure executed in CorelDRAW.

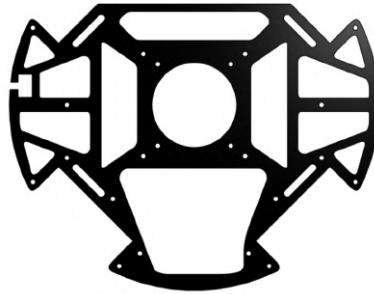


Fig. 2. Extruded and optimized 3D chassis model in the Onshape cloud environment.

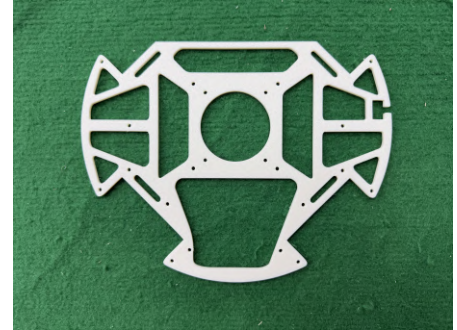


Fig. 3. The final 3D-printed ASA chassis, featuring strategic cutouts for weight reduction.

We manufactured our custom components using 3D-printed ASA (Acrylonitrile Styrene Acrylate), specifically selecting it over common alternatives like PLA and PETG, which lack the requisite balance of surface hardness and impact toughness for aggressive matches. However, initial physical testing revealed a critical paradox: the parts were simultaneously overweight and structurally weak, flexing under impact. To resolve this, we collaboratively optimized the chassis geometry in our shared Onshape workspace. Applying a deep understanding of mechanical stress distribution, we reduced mass through strategic cutouts in non-load-bearing regions. Simultaneously, we mitigated stress concentrations by engineering calculated fillets and reinforcing ribs into high-stress joints, allowing the structure to effectively dissipate kinetic energy. Finally, we fine-tuned our slicer parameters—prioritizing denser outer wall perimeters over heavy internal infill, and optimizing the print orientation to ensure layer adhesion lines did not align with impact vectors. This comprehensive optimization successfully yielded a highly durable, rigid robot well under the strict competitive weight limit.

B. Kicker

1) *Electromagnetic Kicking Mechanism & Hardware Iteration:* Our kicking system is designed to deliver maximum transferable kinetic energy to the newly standardized orange golf ball. To achieve this, we custom-designed a large, ultra-low ground clearance striking plate in Autodesk Fusion 360 that interacts with a larger portion of the ball's spherical surface. The core mechanical actuator driving this plate underwent a crucial physical upgrade to meet our power requirements.

Iteration 1: EML-0826B Solenoid

In our early prototypes, strict spatial constraints at the front of the chassis restricted us to using the compact EML-0826B solenoid. While its small form factor was easy to integrate mechanically, practical field testing revealed a severe limitation: its mechanical stroke force was completely insufficient to propel the heavier golf ball to competitive distances.

Final Solution: EML-1039B Solenoid Upgrade

To resolve this lack of kinetic power, we completely redesigned the front section of our 3D-printed ASA chassis. By optimizing the internal component layout and slightly shifting the dribbler axis, we successfully expanded the internal volume allocated for the kicking module. This mechanical redesign allowed us to upgrade to the vastly larger and more powerful EML-1039B solenoid. Boasting a robust 35N force rating and a longer mechanical stroke, this upgraded actuator delivers explosive physical power. Our custom Fusion 360 striking plate attaches directly to its reinforced plunger, resulting in highly stable and devastatingly fast kicks.

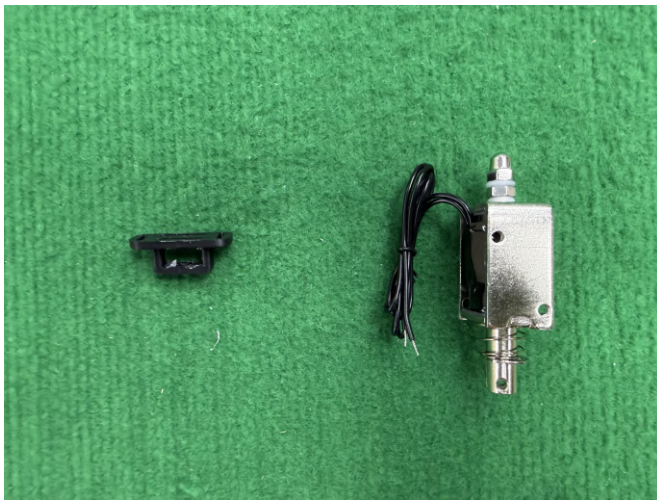


Fig. 4. Iteration 1: The compact EML-0826B solenoid, which lacked sufficient kinetic force for the new ball.

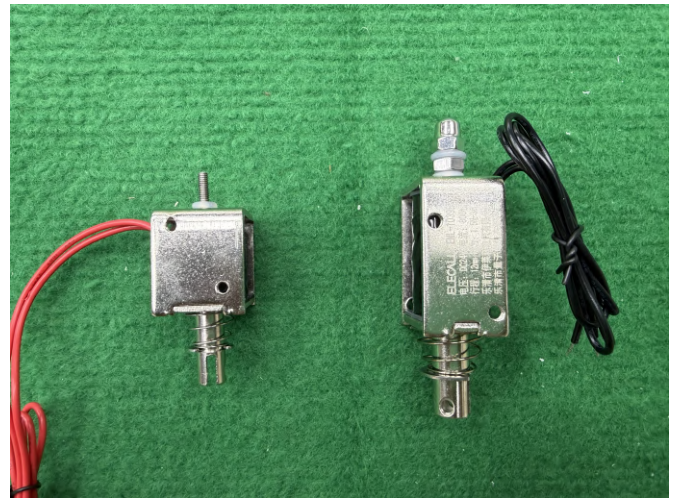


Fig. 5. Hardware comparison illustrating the significant physical size and stroke upgrade from the EML-0826B (left) to our final EML-1039B solenoid (right).

C. Active Dribbler & Dynamic Suspension Iteration

Our active dribbler system is engineered to securely trap the ball against the robot's chassis during rapid holonomic rotations and sudden directional changes. This mechanism underwent rigorous iterative testing to balance motor torque and downward pressure.

Initial Design & Identified Flaws

Initially, we constructed the dribbler using a SPARKHOBBY 1507 3750KV brushless motor paired with a 20A ESC. Power was transferred via a brass coupling and a belt-drive system to a custom 30-degree molded silicone friction roller. This early version featured a fixed-height roller mathematically calculated to perfectly graze the top of the ball.

However, empirical field testing exposed two critical flaws in this early configuration:

1. Insufficient Torque: The high-KV motor (3750KV) lacked the starting torque required to reliably drive the lower shaft under the load of the ball.

2. Lack of Dynamic Pressure: The fixed-height configuration lacked the active downward pressure needed to maintain control. Without dynamic tension, the ball would easily escape the mechanism during aggressive maneuvers or when encountering floor irregularities.

Iterative Solutions & Final Design

To resolve these bottlenecks and achieve maximum stability, we implemented two major upgrades to the dribbler system:

1. Powertrain Upgrade (Addressing Torque): We replaced the initial motor with a T-Motor V2306 V2 KV2400 paired with a 30A ESC. This lower-KV, larger-stator motor provided the robust and reliable torque necessary to drive the roller instantaneously under physical load.

2. Articulated Spring Suspension (Addressing Downward Pressure): To solve the ball retention issue, we completely redesigned the mechanical topology into a split-mechanism articulated dribbler. In this final design, the entire roller and motor assembly pivots on a centralized rear mounting axis, while lateral tension springs apply a continuous downward force. When the ball enters the mechanism, its spherical geometry pushes the roller upward against the spring tension. This mechanical displacement generates a dynamic, constant downward pressure, perfectly locking the ball against the chassis regardless of robot acceleration.

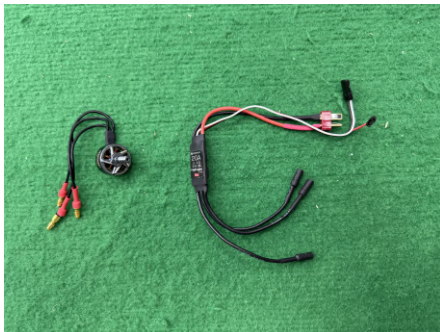


Fig. 6. V1: SPARKHOBBY Motor with fixed-height mount.

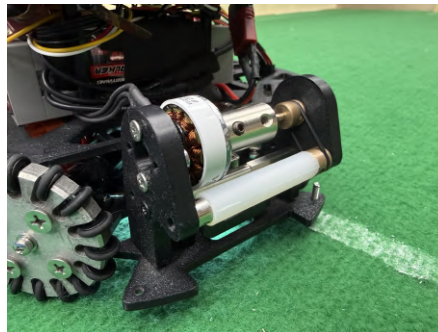


Fig. 7. V2: Upgraded T-Motor V2306 for increased torque.

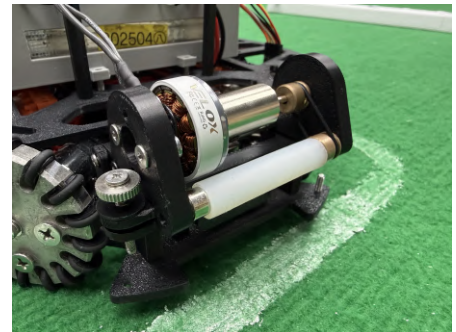


Fig. 8. Final: Articulated split-mechanism with spring suspension.

D. Custom Omnidirectional Wheels

To achieve flawless holonomic kinematics, we engineered our own custom omnidirectional wheels rather than relying on commercial off-the-shelf alternatives. The primary wheel chassis was meticulously designed in our Onshape workspace and subsequently manufactured using precision CNC metal milling. This solid metallic construction ensures absolute structural rigidity and eliminates the lateral flexing issues commonly associated with 3D-printed wheels under high-speed impacts.

Each main wheel assembly consists of 16 independent sub-rollers. To maximize traction and grip on the competition carpet, we utilized Nitrile Butadiene Rubber (NBR) bands as the contact treads for these miniature rollers. Every sub-roller is securely mounted on a rigid steel shaft to minimize rotational friction and mechanical play. Finally, the fully assembled wheels are seamlessly integrated with our Maxon drive motors utilizing custom-machined couplings, the design of which is detailed in the subsequent section.

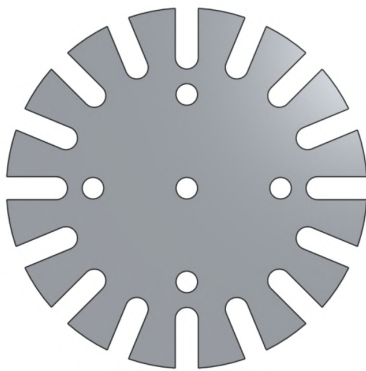


Fig. 9. Onshape CAD design of the omni-wheel base plate.

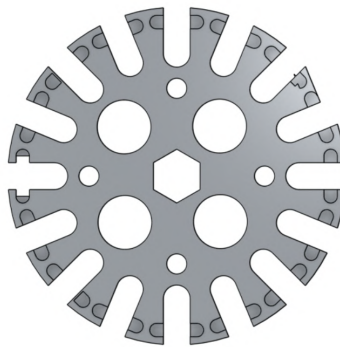


Fig. 10. CAD model detailing the 16 sub-roller mounting slots and hexagonal hub.



Fig. 11. Final CNC-machined wheel assembled with NBR rubber treads and steel shafts.

E. Motor Couplings & Transmission Iteration

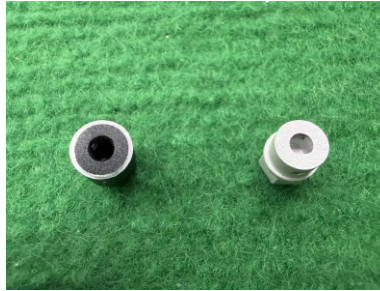


Fig. 12. Evolution of our motor couplings. The final CNC-machined design incorporates a precision D-profile bore to securely lock onto the motor shaft, preventing rotational slippage.

To effectively transfer the torque from our Maxon RE16 motors to the custom omnidirectional wheels, the coupling mechanism underwent several critical iterations to address structural shear failures and rotational slippage.

Iteration 1: 3D-Printed ASA Couplings

Initially, we utilized 3D-printed ASA couplings. While lightweight and easy to rapidly prototype, these printed components lacked the shear strength necessary to withstand the intense, repeated physical collisions typical of RoboCup matches. The internal plastic hubs frequently stripped or fractured under high-torque directional changes.

Iteration 2: CNC Machined Round-Bore Hubs

Seeking enhanced durability, we transitioned to CNC-machined metal couplings. However, our first-generation CNC design featured a standard cylindrical bore secured only by a set screw. This design failed to provide sufficient contact area with the motor's shaft. During sudden accelerations, the set screws would loosen, causing the couplings to slip and physically wear down the motor shafts.

Final Solution: CNC Machined D-Profile Couplings

To definitively solve the slippage issue, we engineered a second-generation CNC coupling featuring a precision-machined D-shaped bore (D-profile). This custom internal geometry perfectly matches the D-shaft of our driving motors, maximizing the mechanical contact area. By physically locking the shaft's flat face against the coupling, this design completely eliminates rotational slipping and ensures optimal torque transfer from the drivetrain to the wheels, even during aggressive defensive collisions.

F. Holonomic Drivetrain & Motor Distribution

To achieve highly responsive omnidirectional maneuverability, our kinematic base utilizes four Maxon RE16 brushed DC motors. These high-efficiency motors were specifically selected for their exceptional torque-to-weight ratio and precise linear response, which are absolutely critical for the rapid acceleration and sudden vector changes demanded by competitive RoboCup matches.

Rather than a traditional differential or orthogonal drive, the four motors are symmetrically distributed around the chassis perimeter at precise 45-degree angles relative to the robot's primary forward axis. This classic "X-drive" holonomic configuration ensures identical kinematic performance vectors along both the lateral and longitudinal axes. It allows the robot to seamlessly strafe, advance, and rotate simultaneously without losing traction or sacrificing velocity.

Mechanically, the motor mounting brackets are continuously integrated into the core 3D-printed ASA chassis structure. To optimize the robot's physical stability against defensive impacts, the motors are nested horizontally and positioned as close to the 22 cm perimeter limit and the ground clearance boundary as possible. This strategic radial mass distribution drastically lowers the robot's overall Center of Gravity (CoG), preventing chassis tilt and ensuring the omnidirectional wheels maintain constant, even pressure against the competition carpet.

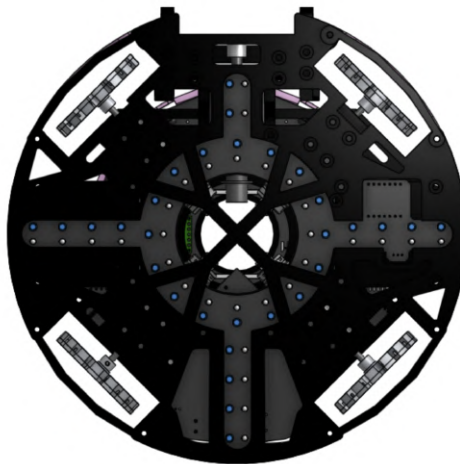


Fig. 13. Bottom view of the chassis illustrating the symmetrical 45-degree holonomic motor distribution, designed to lower the Center of Gravity (CoG) and maximize kinematic efficiency.

III. ELECTRONIC HARDWARE

Our electronic architecture is designed around a distributed processing paradigm to eliminate latency and ensure high-speed control loops. Figure 14 illustrates the overall hardware topology, highlighting the power distribution and serial communication pathways between our microcontrollers and peripherals.

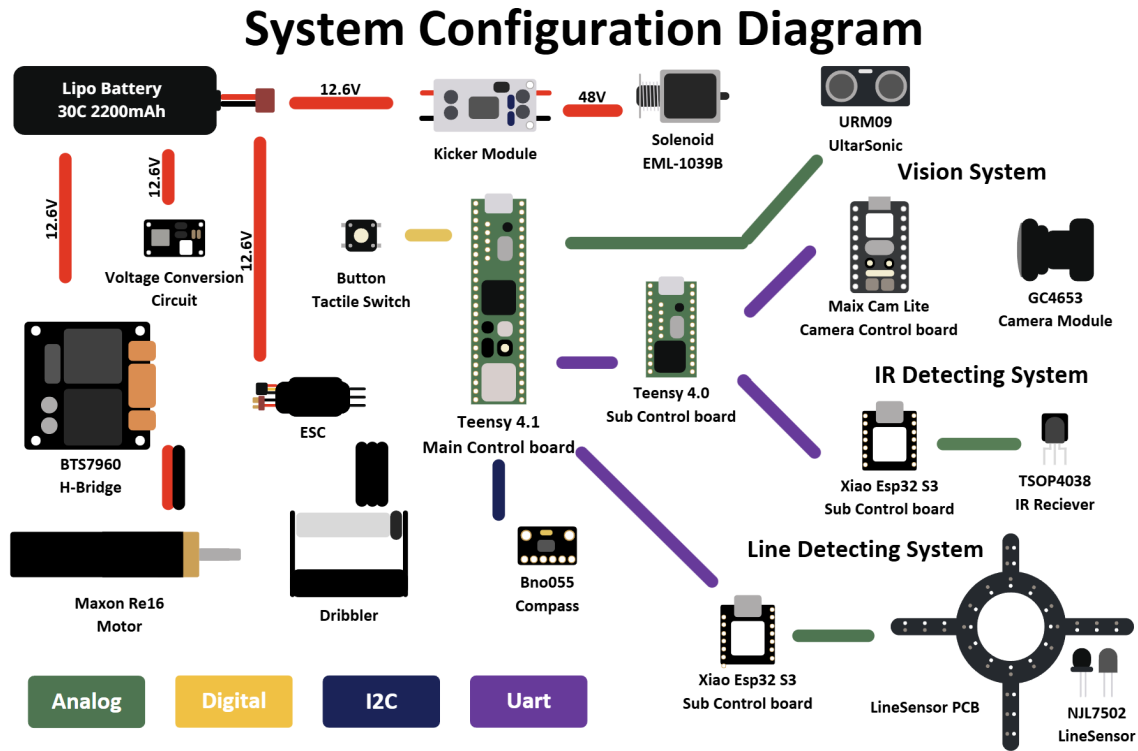


Fig. 14. Hardware Block Diagram detailing the triple-MCU data flow and isolated power routing.

A. Main PCB & Central Controller Iteration

The main printed circuit board (PCB) serves as the central nervous system of our robot. As our computational demands and sensor arrays expanded, our mainboard underwent an extensive five-generation redesign process to optimize spatial efficiency, power delivery, and processing topology.

Iteration 1: MEGA2560 PRO & Arduino Nano

Our initial prototype utilized a simple 8-bit architecture, pairing a MEGA2560 PRO for core logic with an Arduino Nano for sensor polling. While easy to program, this setup lacked the clock speed and memory required for high-frequency holonomic control loops.

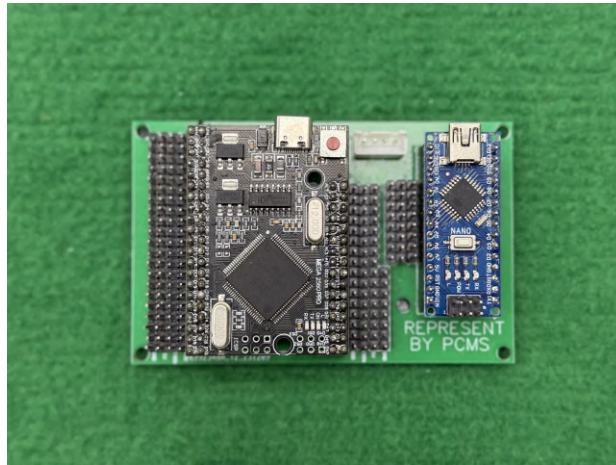


Fig. 15. Iteration 1: The foundational MEGA2560 PRO and Arduino Nano setup.

Iteration 2: Teensy 4.1 & ESP32-S3 Transition

To overcome processing bottlenecks, we completely overhauled the architecture, upgrading to the 32-bit Teensy 4.1 paired with a Seeeduino XIAO ESP32-S3 coprocessor. This generation introduced a dedicated 3.3V step-down buck converter for stable logic power and transitioned to standard pin headers (XH/JST) for reliable, modular sensor connections.

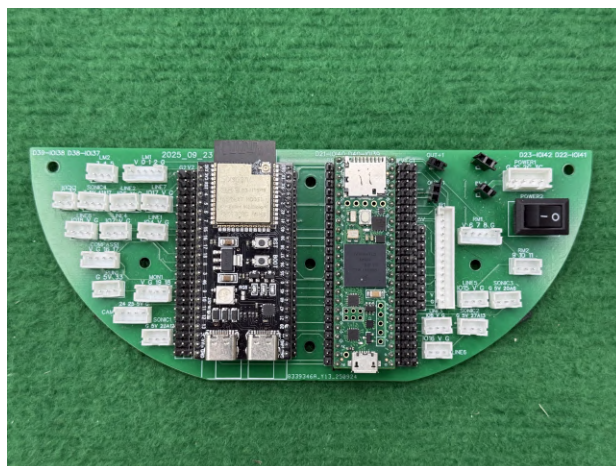


Fig. 16. Iteration 2: Upgraded 32-bit architecture featuring Teensy 4.1, ESP32-S3, and a 3.3V step-down converter.

Iteration 3: Spatial Reallocation & BNO055 Integration

In our third iteration, we mechanically relocated the line detection coprocessor directly onto the bottom sensor PCB to reduce wire clutter. This freed up significant real estate on the mainboard, allowing us to integrate the BNO055 IMU directly onto the main PCB. However, this layout resulted in poor spatial utilization, leaving large areas of the board completely empty.

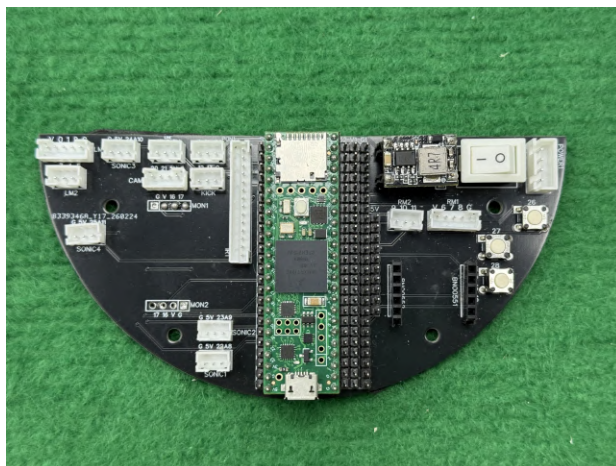


Fig. 17. Iteration 3: The BNO055 was integrated, but the removal of the line coprocessor left the PCB with inefficient empty space.

Iteration 4: Feature Expansion & Redundancy

Attempting to utilize the empty space, we reorganized the circuit traces to include dedicated headers for an external communication module, an OLED display interface, and a digital voltmeter. However, practical field testing revealed that the voltmeter was functionally redundant during gameplay, prompting us to abandon these unnecessary additions in favor of a lighter footprint.

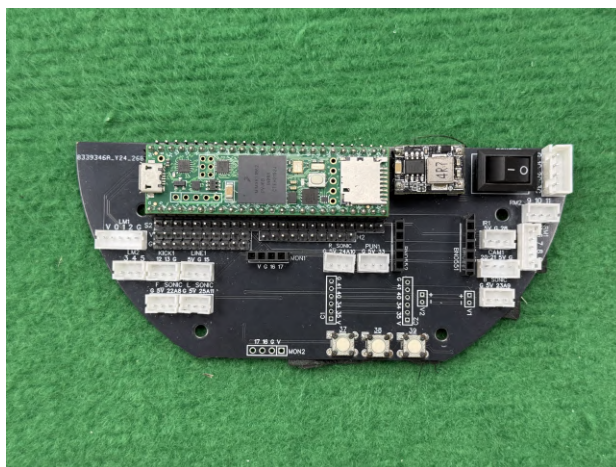


Fig. 18. Iteration 4: Experimental layout featuring headers for OLED, communications, and a voltmeter, which were later deemed redundant.

Final Solution: Extreme Compression & 5A Power Upgrade

For our tournament-ready mainboard, we aggressively compressed the PCB footprint by stripping away all redundant features. Crucially, the integration of our quad Maix Cam Lite array significantly increased the system's overall power draw. To ensure absolute stability and prevent brownouts during sudden motor accelerations, we replaced the original 3A step-down module with a robust 5A buck converter. The final board is highly compact, strictly containing only essential logic routing, IMU headers, and a high-capacity power distribution network.

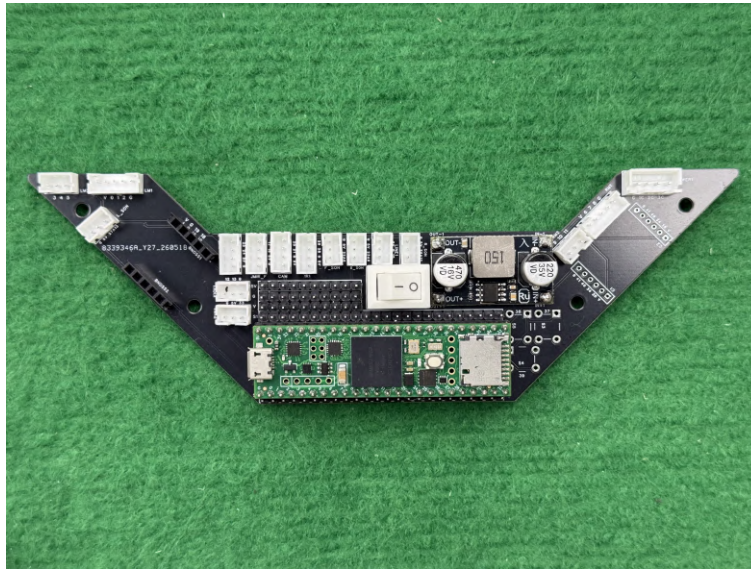


Fig. 19. Final Main PCB: A highly compressed, V-shaped footprint featuring the upgraded 5A buck converter to reliably power the quad-camera neural network.

B. Power Distribution Strategy

As illustrated in our hardware block diagram, the robot is powered by an unregulated 3S LiPo battery supplying a nominal 12.6V. To ensure system stability and isolate sensitive logic components from electromagnetic noise generated by the drivetrain, we implemented a strict multi-rail power distribution network. The raw 12.6V rail is routed directly to the motor driver modules to maximize available torque. Simultaneously, a dedicated DC-DC step-down buck converter regulates the battery voltage to a stable 5V, providing clean logic power to our multi-MCU network (Teensy 4.1, Teensy 4.0, and dual ESP32-S3 modules) and all associated sensors. Finally, a specialized boost converter steps the 12.6V rail up to 48V, charging the high-capacity capacitors required for our EML-1040B solenoid kicker system.

C. Distributed Processing Architecture

We abandoned the traditional single-MCU approach in favor of a highly specialized quad-core network communicating via high-speed Serial interfaces:

- **Master Kinematic Controller (Teensy 4.1):** Serving as the central brain, this MCU directly reads the I2C IMU and the four motor encoders to compute the closed-loop PID holonomic kinematics. It drives the motors, manages the UI display, and executes high-level state machines based on finalized telemetry received from the coprocessors.
- **Vision Aggregator (Teensy 4.0):** Dedicated entirely to image processing, it interfaces simultaneously with four Maix Cam Lite modules, synthesizing 360-degree vision data into unified targeting vectors to offload heavy string-parsing from the main controller.
- **Dual Sensor Coprocessors (2x ESP32-S3):** To prevent analog-to-digital (ADC) polling bottlenecks, we divided our high-density analog arrays between two dedicated Seeeduno XIAO ESP32-S3 modules. The first ESP32-S3 is strictly assigned to infrared tracking, routing 12 hardware-filtered TSOP4038 IR receivers through a single multiplexer. The second ESP32-S3 serves as the boundary manager, rapidly sampling 30 NJL7502L line sensors routed through dual multiplexers, allowing it to instantly flag out-of-bounds states via priority interrupts.

D. High-Current Motor Control

To drive our four Maxon RE16 brushed DC motors, we utilized two commercial WAYNESTARK V3.0 Dual-Channel high-current motor driver modules rather than integrating bare driver ICs onto our custom mainboard. We selected these specific dual-channel modules because their massive current rating provides extreme thermal headroom, ensuring the drivers will never overheat or fail during the aggressive acceleration and sudden directional changes that occur in RoboCup matches.

These two driver boards are connected directly to our unregulated 12.6V battery rail to supply maximum available torque to all four motors. To dictate movement, our Teensy 4.1 microcontroller sends 3.3V logic signals—specifically, high-frequency Pulse Width Modulation (PWM) and digital direction states—to the dual inputs of each motor channel. The WAYNESTARK modules utilize robust internal H-bridge circuits to amplify these low-power logic signals, rapidly switching the high-current 12.6V power to the motors.

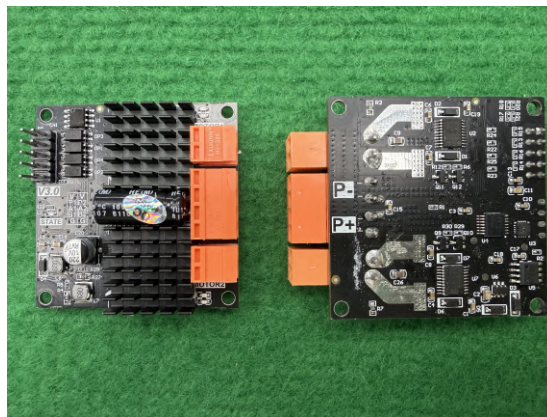


Fig. 20. WAYNESTARK V3.0 Dual-Channel H-Bridge Driver Module, featuring integrated heat sinks for high-current thermal management.

By adjusting the duty cycle of the PWM signals, our software dynamically scales the voltage applied to the wheels, achieving the highly granular speed control required for precise holonomic kinematics. Furthermore, utilizing these high-grade commercial modules provides essential built-in hardware protection, seamlessly isolating our sensitive 3.3V logic systems and the Teensy microcontroller from the harsh inductive voltage spikes (back-EMF) generated by the DC motors.

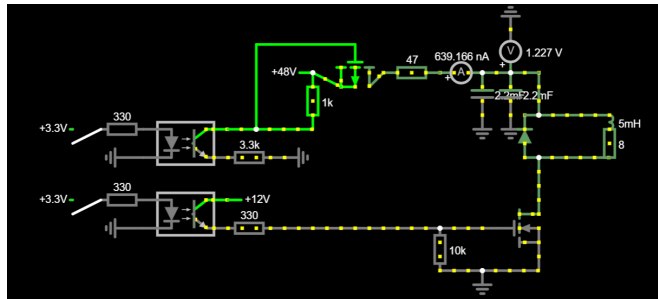
E. Electromagnetic Kicker Circuit & Simulation

Driving the 48V solenoid kicking system within the spatial constraints of the new competition rules presented a significant electronic power management challenge.

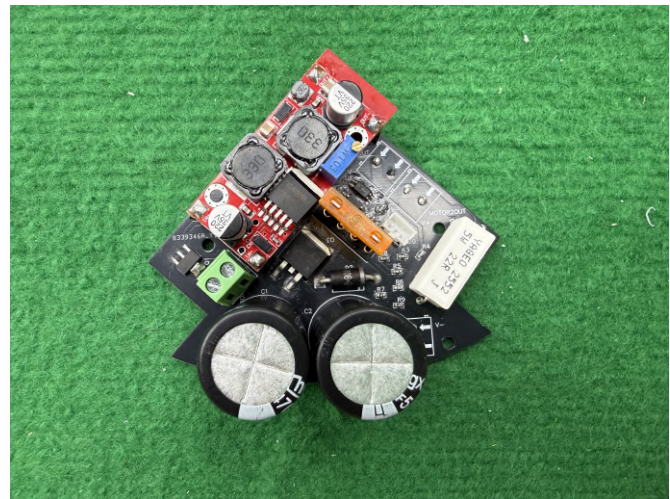
Iteration 1: Direct Drive & Initial Boost PCB

Our initial approach utilizing a 12V direct-drive electromagnet yielded a kicking distance of only 5 cm. Seeking to maximize instantaneous current output, we engineered our first custom Kicker PCB featuring a voltage boost module and a capacitor bank. However, upon testing, the output voltage plummeted to 0V instead of the target 48V.

To diagnose this critical failure, we reconstructed the entire mechanism in the *CircuitJS* simulation environment. The simulation revealed that the massive inrush current required to initially charge the completely empty capacitors was triggering the boost converter's internal overcurrent protection, forcing a fail-safe shutdown.



(a) Circuit Schematic



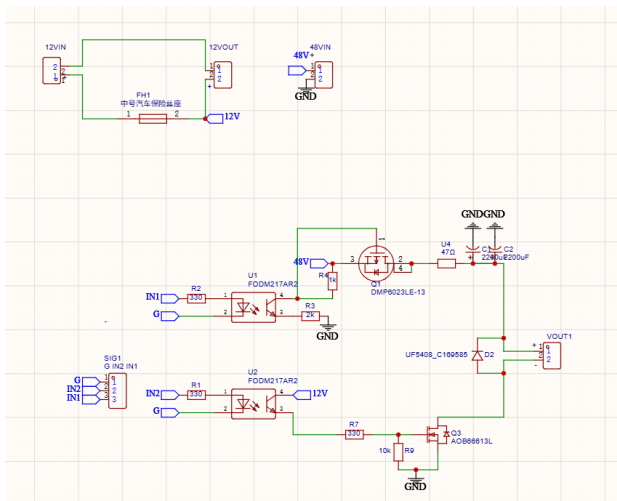
(b) Assembled PCB

Fig. 21. Iteration 1 Kicker System: Initial schematic and physical PCB, which suffered from inrush current failure.

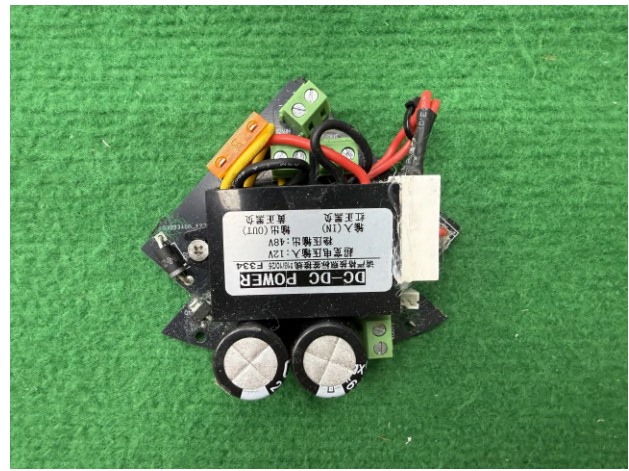
Final Solution: Dual-Channel Isolated Architecture

Armed with the simulation data, we successfully redesigned our second-generation Kicker PCB. We introduced a 47Ω current-limiting series resistor between the boost converter and the capacitor bank, added a safety fuse, and upgraded to a highly robust 48V 2A boost module capable of handling the 1.1A initial surge calculated in our simulation.

Our final system features a dual-channel 48V drive circuit with complete optocoupler isolation to protect the sensitive 3.3V logic of our Teensy 4.1 master controller. The upper logic channel drives an optocoupler which switches a BJT transistor and a P-channel power MOSFET. The lower logic channel drives a second optocoupler connected to an N-channel MOSFET with a $10k\Omega$ pull-down resistor. When triggered synchronously, this isolated H-bridge topology rapidly dumps the stored energy from two parallel 2.2mF capacitors into the 12V 35N electromagnet. A freewheeling diode safely suppresses the high-voltage inductive back-EMF, ensuring extreme reliability.



(a) Isolated Dual-Channel Schematic



(b) Assembled PCB

Fig. 22. Final Kicker System: Redesigned schematic and physical PCB featuring the upgraded 48V boost module, 47Ω current-limiting resistor, and optical isolation.

F. Sensors and Measuring Units

1) *Vision Aggregation System & Hardware Iteration:* Our current 360-degree environmental awareness system is the result of a rigorous hardware iteration process aimed at eliminating optical distortion and processing bottlenecks.

Iteration 1: Catadioptric OpenMV System

Our initial approach utilized an OpenMV H7 paired with a custom heated silica-based convex mirror to achieve 360-degree vision via a catadioptric reflection system (Fig. 23). However, physical testing revealed fatal flaws: the curved mirror introduced severe geometric distortion, and traditional color-thresholding algorithms proved highly unreliable under the dynamic lighting conditions of tournament environments, leading to frequent false-positive detections.

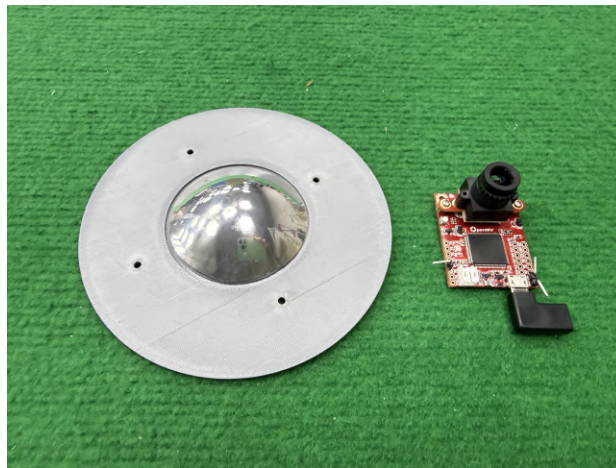


Fig. 23. Initial Catadioptric Vision System utilizing OpenMV H7 and a custom silica-based convex mirror, which suffered from severe optical distortion.

Iteration 2: Evaluating the M5Stack UnitV2

Seeking a more robust, neural-network-based approach, we evaluated the M5Stack vision module, a popular choice among many competitor teams. While it provided AI-driven object detection, our benchmarking revealed that its AI recognition efficiency and overall framerate were insufficient to keep pace with the rapid holonomic kinematics required by our control loops.



Fig. 24. Testing the M5Stack module, which lacked the required framerate for high-speed kinematics.

Iteration 3: Quad Maix Cam Array & Power Bottleneck

Inspiration for our neural-network architecture came from observing our school's RoboCup Rescue team utilizing the Sipeed Maix Cam. To strictly adhere to the 1.5 kg Soccer Lightweight constraint, we sourced

the Maix Cam Lite and initially deployed an array of four modules, each retrofitted with a 130° FOV M12 mount wide-angle lens. While this quad-camera topology completely eliminated blind spots, full-match simulations revealed a fatal power consumption issue. The continuous multi-core NPU processing of four simultaneous cameras caused our 2200mAh 3S LiPo battery voltage to plummet from a fully charged 12.6V down to a critical 11.9V–12.0V threshold in just five minutes. This severe voltage drop threatened to destabilize the motor drivers and logic circuits, making it impossible to sustain a standard match duration.

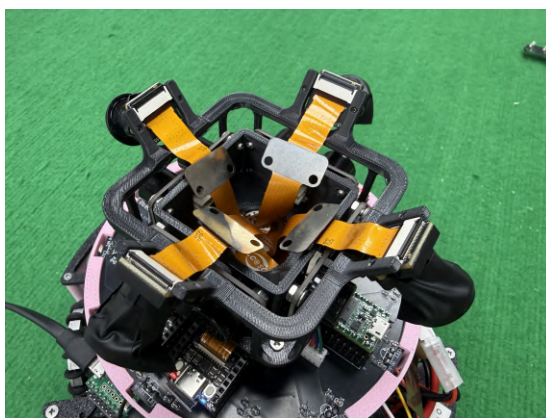


Fig. 25. Iteration 3: The initial quad-camera array. While it provided complete 360-degree coverage, its high power consumption caused severe battery voltage drops.

Final Solution: Dual-Axis Vision Topology

To resolve this critical power bottleneck, we rigorously re-evaluated our tactical vision requirements. Empirical data from field tests showed that the lateral (left and right) cameras had a significantly lower utilization rate during holonomic gameplay compared to the longitudinal axes. We optimized our system by removing the lateral modules, finalizing a dual-camera (front and rear) symmetrical architecture. This strategic reduction drastically lowered overall power consumption and top-heavy mass. Coupled with our Teensy 4.0 Vision Aggregator, this streamlined setup still provides highly efficient dual-hemispheric field awareness while ensuring the robot maintains stable voltage and peak performance throughout an entire competitive match.

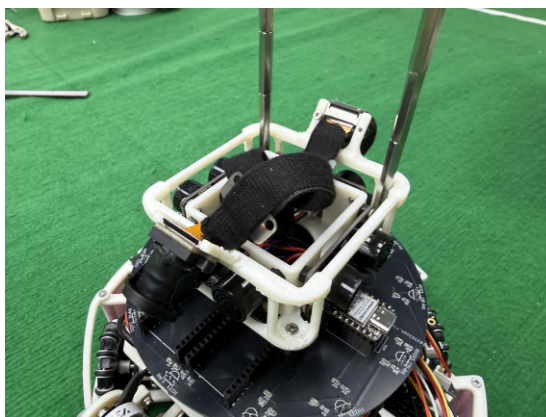


Fig. 26. Final Dual-Axis Vision Topology: A streamlined front-and-rear Maix Cam Lite configuration that perfectly balances wide field awareness with sustainable battery life.

2) *Infrared Tracking & Hardware Filtering Iteration:* Similar to our vision system, our infrared ball-tracking architecture underwent significant generational upgrades to adapt to changing competition regulations and computational demands.

Iteration 1: Joinmax JMP-BE-1732 Array

During the era of the legacy RCJ05R pulsed infrared ball, our first-generation robot utilized Joinmax JMP-BE-1732 modules. We mounted two of these sensors (front and rear) and read their intensity values via a UART interface to determine basic distance and orientation. However, this setup suffered from a severely limited detection range. Furthermore, these modules were obsolete, leaving us with no replacement parts in the event of hardware failure and prompting a complete architectural redesign.

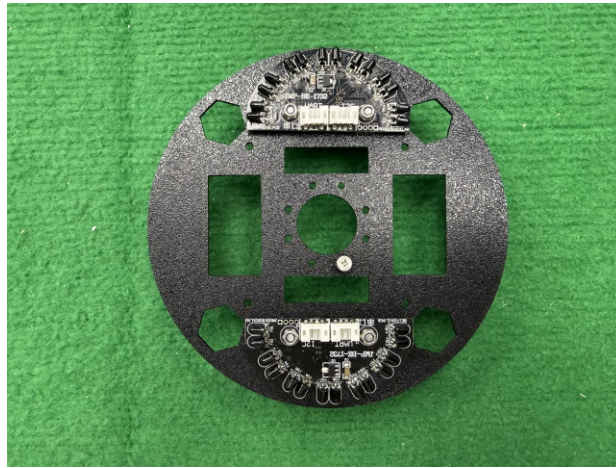


Fig. 27. Iteration 1: Legacy Joinmax JMP-BE-1732 UART sensor setup, which suffered from limited range and obsolescence.

Iteration 2: 10cm TSSP58038 Ring (Direct to Teensy 4.1)

Seeking higher precision, we designed a custom 10 cm diameter PCB ring featuring 12 TSSP58038 IR receivers. In this architecture, all 12 sensor outputs were wired directly to the digital pins of our Teensy 4.1 master controller. While these sensors provided excellent proportional intensity values to calculate the ball's angle, we quickly discovered a fatal flaw: continuously polling and processing 12 high-frequency signals consumed excessive CPU cycles, severely crippling the Teensy's kinematic control loop frequency.



Fig. 28. Iteration 2: Custom 10cm PCB ring with 12 TSSP58038 sensors, initially designed to connect directly to the Teensy 4.1 master controller.

Iteration 3: Coprocessor & Multiplexer Integration

To resolve the critical processing bottleneck identified in Iteration 2, we completely decoupled the IR polling from the main controller. We introduced a Seeeduno XIAO ESP32-S3 as a dedicated IR coprocessor paired with a hardware multiplexer. The ESP32-S3 perfectly handled the intensive sensor math, streaming only the finalized trajectory vectors to the Teensy without any performance penalty.

Final Solution: TSOP4038 Array, Hardware Filtering & Repositioning

When the RoboCup Technical Committee updated the rules to adopt the new orange IR golf ball, our TSSP58038 sensors could no longer acquire valid data due to the ball's different emission characteristics. After extensive research, we selected the TSOP4038 receiver. Because the TSOP series is typically designed for digital toggling rather than analog intensity measurement, we engineered a custom analog Low-Pass Filter (LPF) circuit to integrate the incoming pulses, successfully extracting a clean, proportional intensity voltage for the new golf ball. We maintained our highly successful 12-sensor multiplexed architecture but made a crucial mechanical optimization: we relocated the entire IR array from the top of the PCB to the bottom. This lowered the sensors' perspective, significantly expanding the effective detection range and preventing the robot's own chassis from casting infrared shadows.

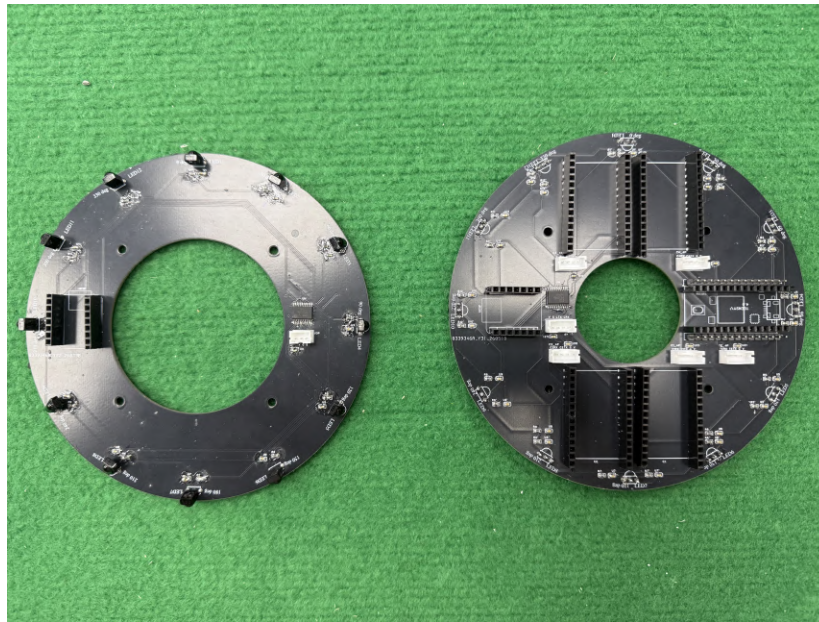
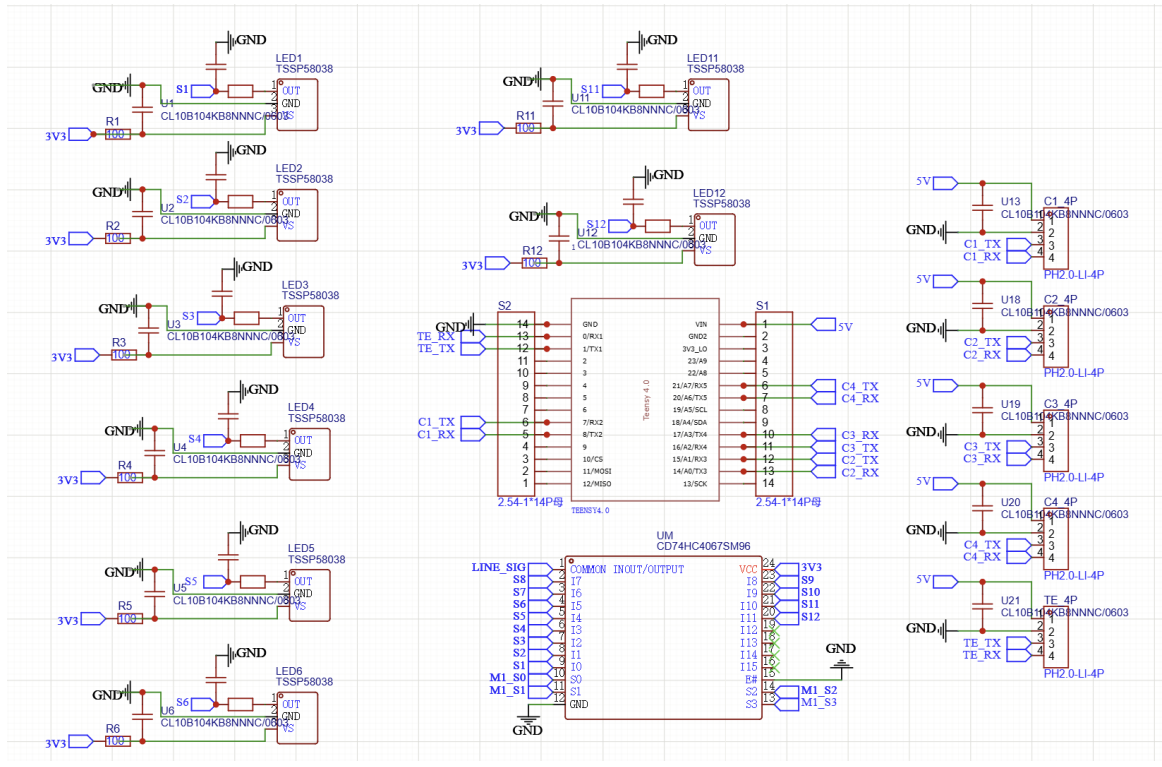


Fig. 29. Comparison of our IR PCBs: Iteration 2 (top-mounted TSSP58038) and the Final Solution (bottom-mounted TSOP4038 with custom LPF circuitry).



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Fig. 30. Schematic diagram of the final infrared array, detailing the custom analog Low-Pass Filter (LPF) circuitry applied to each receiver and the hardware multiplexer integration.

3) *High-Density Line Detection Array & Hardware Iteration:* Similar to our vision and infrared systems, our boundary detection hardware underwent multiple iterations to achieve the high density and processing speed required for advanced vector calculations.

Iteration 1: Joinmax BE-1413 & Arduino Nano

Our first-generation line detection system utilized four Joinmax BE-1413 photoresistor sensors located at the corners of the robot, processed by an Arduino Nano using hardware interrupts. While functional for basic out-of-bounds detection, this four-point setup offered extremely low resolution.

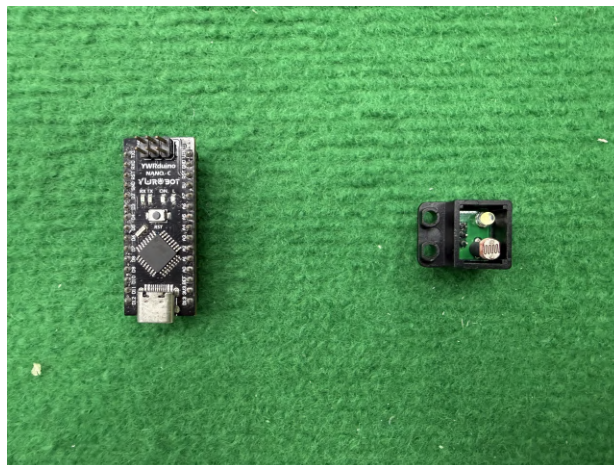


Fig. 31. Iteration 1: Arduino Nano and Joinmax BE-1413 photoresistor sensor.

Iteration 2: HJ517 Grayscale Modules & ESP32-S3

To improve processing speed and accuracy, we upgraded to an ESP32-S3 microcontroller and six HJ517 grayscale sensor modules. Although the response time improved significantly, we realized that six sensors were still too sparsely distributed to perform advanced algorithmic calculations, such as determining the exact angle and geometric vector of the white line relative to the robot.

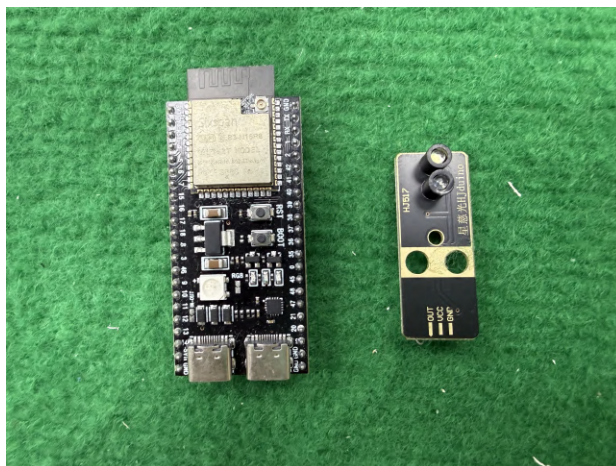


Fig. 32. Iteration 2: ESP32-S3 paired with HJ517 grayscale modules.

Iteration 3: 23-Sensor NJL7502L Custom Array

Moving away from commercial modules, we designed a custom PCB array utilizing 23 NJL7502L photodiodes paired with custom LED illumination. We implemented voltage divider resistor circuits and two hardware multiplexers (CD74HC4067SM96) to route all analog signals into a Seeeduino XIAO ESP32-S3 coprocessor. However, empirical testing revealed a critical flaw: the physical gaps between the 23 sensors were just large enough that the white boundary line could occasionally slip through undetected, depending on the robot's angle of approach.

Final Solution: 30-Sensor High-Density Array

To completely eliminate these physical blind spots, we increased the sensor density by adding 7 additional NJL7502L sensors and LEDs, bringing the total to 30. Using the exact same dual-multiplexer ESP32-S3 architecture, this high-density layout ensures the white line is always detected by at least one sensor. Furthermore, this extreme resolution allows our coprocessor to pinpoint exactly which specific sensors are triggered, instantly calculating the precise geometric vector and angle of the boundary line to feed into our active vector containment algorithms.

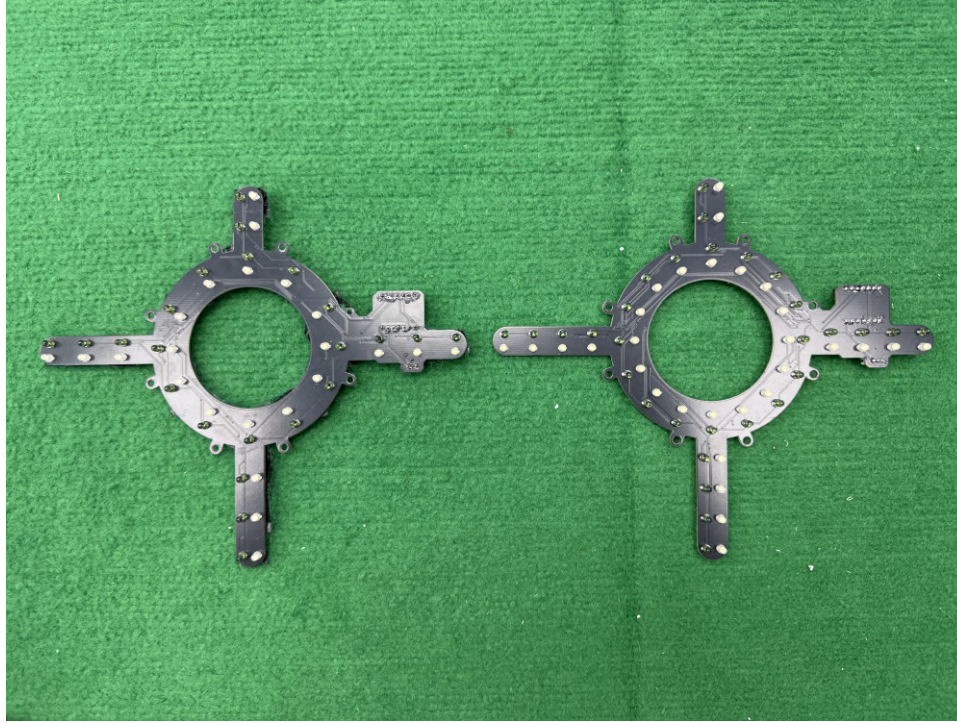


Fig. 33. Comparison of our custom line sensor PCBs. Left (Iteration 3): The 23-sensor array which suffered from physical blind spots. Right (Final Solution): The upgraded high-density 30-sensor array.

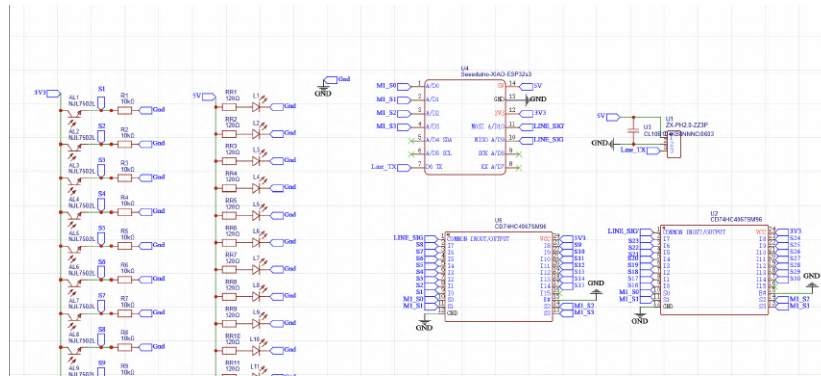


Fig. 34. Schematic diagram of the final 30-sensor array, detailing the Sceduino XIAO ESP32-S3, dual CD74HC4067SM96 multiplexers, and the NJL7502L voltage divider circuits.

4) *Distance Measurement & Ultrasonic Sensor Iteration:* Precise spatial localization is critical for our robot's field positioning and obstacle avoidance. Our ultrasonic hardware underwent a necessary evolution to balance measurement precision, refresh rate, and component availability.

Iteration 1 & 2: Legacy Joinmax & US-016 Modules

Our first-generation system utilized the Joinmax JMP-BE-6302, which provided highly precise analog distance signals. However, its long-term viability was compromised because it had been discontinued years ago, making replacement parts impossible to source. As a direct substitute, we transitioned to the US-016 module in our second iteration. Unfortunately, its performance fell significantly short of our kinematics requirements. During field testing, the US-016 exhibited severe data fluctuations and reading

errors at distances exceeding 30 cm, leading us to abandon it after only a few months of use.

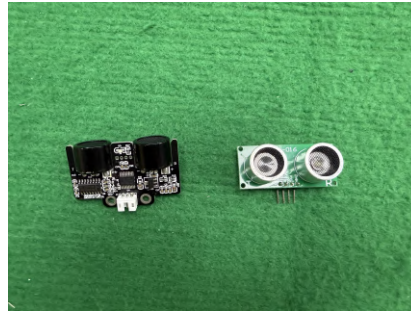


Fig. 35. Left (Iteration 1): The obsolete Joinmax JMP-BE-6302. Right (Iteration 2): The US-016 module, which suffered from severe measurement inaccuracies beyond 30 cm.

Final Solution: DFRobot URM09 Array

For our final architecture, we conducted further market research and integrated the DFRobot URM09 ultrasonic sensor. At a highly cost-effective price point of approximately \$5 per unit, it perfectly fulfilled our rigorous demands. It boasts a rapid 30Hz refresh rate and a highly reliable measurement range from 2 cm up to 500 cm. To achieve comprehensive 360-degree environmental awareness, we deployed four URM09 sensors on each robot—oriented in the four cardinal directions—ensuring robust, high-speed wall-distance verification and spatial mapping.



Fig. 36. Final Solution: The DFRobot URM09, offering a high-speed 30Hz refresh rate and a 2-500cm range. Four units are deployed per robot for cardinal spatial awareness.

IV. ROBOT'S GENERAL LOGIC AND BEHAVIOR

A. Distributed Sensor Data Processing

To maintain an uninterrupted, high-frequency kinematic control loop on our Teensy 4.1 master controller, we offloaded all intensive analog polling and preliminary algorithmic synthesis to our dual Seeeduno XIAO ESP32-S3 coprocessors. This distributed architecture ensures that the central brain only receives mathematically smoothed, finalized telemetry via high-speed UART, completely eliminating processing latency.

1) *Top-5 Infrared Vector Synthesis:* Our custom infrared ring utilizes 12 hardware-filtered TSOP4038 receivers. However, relying solely on the single sensor with the highest analog intensity often results in trajectory jitter due to ambient noise, floor reflections, and signal decay. To achieve absolute 1-degree tracking resolution, our dedicated IR coprocessor executes a custom Top-5 Vector Synthesis algorithm.

The algorithm continuously polls all 12 multiplexed channels and dynamically isolates the five receivers with the highest analog intensity (S_i). To exponentially prioritize the most direct infrared emissions and suppress peripheral noise, we apply quadratic weighting to these top five values. The finalized trajectory vector is then synthesized using trigonometric summation based on each sensor's fixed physical angle (θ_i):

$$V_x = \sum_{i=1}^5 (S_i)^2 \cos(\theta_i)$$

$$V_y = \sum_{i=1}^5 (S_i)^2 \sin(\theta_i)$$

$$\theta_{raw} = \arctan 2(V_y, V_x)$$

To eliminate any residual hardware jitter, the raw angle is processed through an Exponential Moving Average (EMA) filter with a smoothing factor of $\alpha = 0.30$. This ensures the Teensy 4.1 receives a flawlessly smooth, zero-latency tracking vector, empowering stable holonomic pursuits even during aggressive rotational maneuvers.

2) *High-Density Boundary Vector Calculation:* Detecting the white out-of-bounds line with 30 closely packed NJL7502L photodiodes presents significant data processing challenges. To prevent the Teensy 4.1 from being overwhelmed by analog-to-digital (ADC) conversions, we designed a decentralized boundary detection architecture utilizing an ESP32-S3 coprocessor and high-speed (921600 baud) UART telemetry.

To counteract ambient lighting variations and hardware tolerances across the 30 sensors, the ESP32-S3 utilizes a dynamic calibration sequence. During initialization, the robot scans the field for eight seconds to record the absolute maximum (white line) and minimum (green carpet) analog values for each individual sensor. The coprocessor calculates and stores a unique localized midpoint threshold for each photodiode in its non-volatile memory. During gameplay, the ESP32-S3 instantly binarizes the real-time analog readings against these custom thresholds, compiling a highly compressed 30-bit string (representing active and inactive states) to stream to the master controller.

Upon receiving the 30-bit telemetry array, the Teensy 4.1 computes the exact geometric normal vector of the boundary line. Instead of executing complex conditional logic, our software utilizes Unit Vector Superposition. For each active sensor ($B_n \in \{0, 1\}$), the algorithm extracts its fixed mechanical mounting angle (θ_n) and calculates the cumulative force vectors (F_x, F_y) along the chassis axes:

$$F_x = \sum_{n=1}^{30} B_n \cos(\theta_n)$$

$$F_y = \sum_{n=1}^{30} B_n \sin(\theta_n)$$

$$\Phi_{line} = \arctan 2(F_y, F_x)$$

This trigonometric synthesis translates discrete sensor triggers into a singular, highly precise line angle (Φ_{line}). Furthermore, the master controller implements a time-gated out-of-bounds state machine. Upon initial line contact, the software records a timestamp and the baseline angle. If the robot's subsequent trajectory deviates beyond a strict angular threshold within a 700 ms window, the state machine locks the robot into an active escape maneuver, forcing a closed-loop holonomic retreat precisely 180° opposite to the initial impact vector until the chassis is fully clear of the boundary.

B. Closed-Loop Kinematics & Holonomic PID Control

To execute the finalized trajectory vectors calculated by the coprocessors, the Teensy 4.1 master controller operates a high-frequency kinematic loop. Our drivetrain features a symmetrical 45-degree X-drive configuration powered by four Maxon motors. Rather than relying on simple directional toggles, we implemented a Field-Centric closed-loop control system to ensure fluid, omnidirectional maneuverability independent of the robot's physical orientation.

1) *Field-Centric Drive & Kinematic Matrix*: To maintain absolute spatial awareness, the main controller continuously polls the onboard BNO055 IMU. The software calculates the relative robot angle (θ_{robot}) by subtracting the current IMU heading (θ_{imu}) from the absolute target movement angle (θ_{target}). The desired translational speed is then decomposed into lateral (V_x) and longitudinal (V_y) vectors using basic trigonometry:

$$V_x = \text{Speed} \cdot \sin(\theta_{robot})$$

$$V_y = \text{Speed} \cdot \cos(\theta_{robot})$$

Simultaneously, a closed-loop Proportional-Integral-Derivative (PID) controller calculates the rotational velocity (ω) required to maintain a locked heading, compensating for defensive impacts or uneven carpet friction. The PID equation isolates the heading error $e(t)$ normalized to a $[-180^\circ, 180^\circ]$ domain:

$$\omega = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

These three dynamic variables (V_x, V_y, ω) are mathematically superimposed and translated into localized Wheel Velocities (W) for the Front-Left (FL), Back-Left (BL), Front-Right (FR), and Back-Right (BR) motors using the following inverse kinematics matrix:

$$\begin{bmatrix} W_{FL} \\ W_{BL} \\ W_{FR} \\ W_{BR} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ \omega \end{bmatrix}$$

2) *Dynamic Proportional Vector Scaling*: A common critical failure in holonomic robotics occurs when the mathematical superposition of V_x , V_y , and ω exceeds the hardware's maximum Pulse Width Modulation (PWM) limit (255). Standard clipping (capping values at 255) physically distorts the intended geometric vector, causing the robot to drift unpredictably during aggressive maneuvers.

To solve this, our control loop implements a Dynamic Proportional Scaling algorithm. Before sending signals to the H-Bridge drivers, the system identifies the absolute maximum calculated wheel velocity (W_{max}). If $W_{max} > 255$, the software scales all four motor outputs proportionally by a factor of $(255/W_{max})$. This crucial mathematical constraint ensures that even when the motors are pushed to their absolute physical limits, the desired kinematic trajectory and rotational heading remain flawlessly intact.

C. Adaptive Offensive State Machine

The primary objective of the attacking robot is to acquire ball possession seamlessly and navigate toward the opponent's goal while actively evading defensive collisions. To achieve this, we developed a highly adaptive offensive state machine that fuses trigonometric trajectory calculations with active spatial awareness.

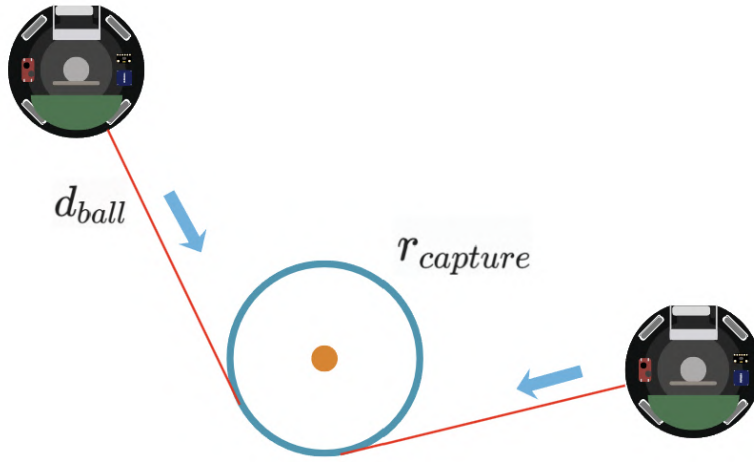


Fig. 37. Geometric visualization of the Tangent Wrap-Around algorithm. The robot calculates a dynamic θ_{offset} utilizing the imaginary $r_{capture}$ to generate a fluid, collision-free interception trajectory.

1) *Tangent Wrap-Around Algorithm*: Traditional chase algorithms often result in the robot colliding directly with the ball, knocking it further away. To solve this, our software calculates an imaginary capture circle (with radius $r_{capture}$) around the ball. Instead of moving directly toward the ball's coordinates, the robot targets a mathematically smoothed point along the tangent of this circle.

By reading the ball's relative distance (d_{ball}) and angle (θ_{ball}) from the perception coprocessors, the Teensy 4.1 calculates a dynamic offset angle (θ_{offset}) using the inverse tangent function:

$$\theta_{offset} = \arctan\left(\frac{r_{capture}}{d_{ball}}\right)$$

To dictate the final movement vector (θ_{chase}), the algorithm employs a piece-wise approach based on the ball's location. If the ball is located directly in front of the robot within a narrow $\pm 15^\circ$ cone, the software applies a $1.5\times$ angular multiplier for a rapid, aggressive frontal interception. For all other angles, the θ_{offset} is dynamically added to or subtracted from the ball's vector:

$$\theta_{chase} = \begin{cases} 1.5 \times \theta_{ball}, & \text{if } |\theta_{ball}| \leq 15^\circ \\ \theta_{ball} \pm \theta_{offset}, & \text{otherwise} \end{cases}$$

This mathematical approach ensures the robot generates a fluid, collision-free trajectory, naturally wrapping around the ball to establish a perfect rear-capture angle.

2) *Invisible Barrier & Dynamic Spatial Evasion*: During aggressive offensive pushes, robots frequently collide with the field boundaries or the goalposts, causing them to drop the ball. To counter this, we implemented an “Invisible Barrier” evasion algorithm utilizing the four DFRobot URM09 ultrasonic sensors.

The algorithm continuously monitors the lateral and longitudinal wall distances. If the robot enters a predefined critical proximity threshold (e.g., $u_{s_{left}} < Limit_{wall}$), the algorithm automatically generates proportional counter-velocities (a_{cx}, a_{cy}) in the opposite direction. These dynamic evasion vectors are superimposed onto the primary holonomic chase matrix:

$$V'_x = V_{x_original} + a_{cx}$$

$$V'_y = V_{y_original} + a_{cy}$$

This creates a virtual repulsion field around the robot. It allows the attacker to maintain aggressive ball possession while seamlessly gliding parallel to walls and safely maneuvering around the goalposts to find the optimal striking gap.

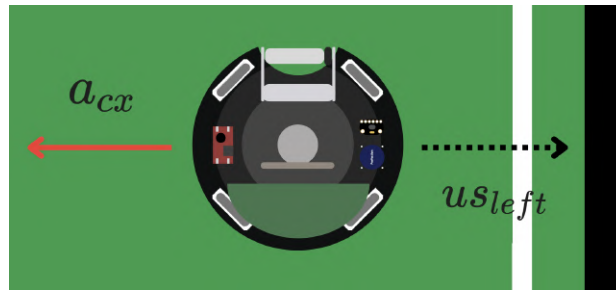


Fig. 38. The Invisible Barrier evasion logic. Ultrasonic sensors detect critical proximity to physical obstacles, generating proportional counter-vectors (a_{cx}, a_{cy}) to maintain possession while searching for the optimal gap-targeted strike.

3) *Gap-Targeted Striking & Tactical Repositioning*: Once the active dribbler secures the ball, the vision system scans the opponent’s goal. If a clear Line of Sight (LoS) is established, the robot aligns its chassis to the goal center. If an enemy defender obstructs the path, the algorithm calculates the largest unguarded spatial gap between the defender and the goalposts, dynamically adjusting the alignment vector before triggering the 48V electromagnetic solenoid for an unblockable shot.

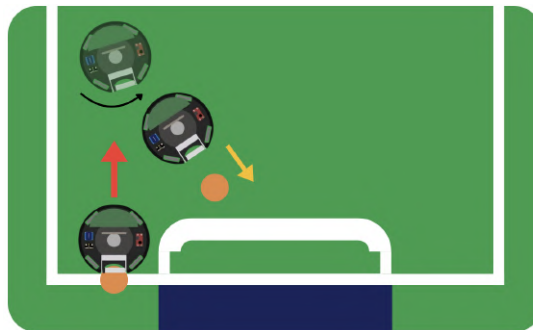


Fig. 39. Tactical Repositioning and Gap-Targeted Striking. When a direct shot is obstructed or the robot is in a suboptimal position, it utilizes the active dribbler to secure the ball, executes a strategic holonomic repositioning maneuver, and pivots to align with the optimal unguarded spatial gap for a clear strike.

D. Dynamic Defensive State Machine

Unlike the attacking unit, the defending robot must prioritize territorial control and shot interception over aggressive ball pursuit. To achieve this, we engineered a Dynamic Defensive State Machine that utilizes multisensor spatial localization to lock the robot into a mathematically constrained defensive perimeter.

1) *Multisensor Spatial Localization*: Before executing any defensive maneuvers, the robot must establish its absolute coordinates on the field. The Teensy 4.1 fuses telemetry from two primary perception systems: the Quad DFRobot URM09 ultrasonic array provides real-time distance metrics to the physical walls ($u_{Sback}, u_{Sleft}, u_{Sright}$), while the Dual Maix Cam array verifies the relative distance and angle to our own goalpost.

By superimposing these datasets, the master controller calculates its longitudinal coordinate (Y_{robot}). Once Y_{robot} falls within the predefined threshold of our penalty area, the system automatically transitions from a generic neutral state into the Dynamic Defensive State, prioritizing goal-line protection.

2) *Active Vector Containment & Holonomic Strafing*: A common critical error for defending robots is accidentally stepping out of the penalty box or crossing the goal line during aggressive blocking, resulting in severe penalty calls. To completely eliminate this, our software utilizes an “Active Vector Containment” protocol.

When the robot operates within the defensive zone, it utilizes the continuous normal vector (Φ_{line}) generated by the 30-sensor high-density line array. Instead of traditional “bounce-back” algorithms, the state machine mathematically locks the robot’s longitudinal velocity axis (V_y). The kinematic matrix is dynamically constrained so that the robot cannot translate backward into the goal or forward out of the penalty box:

$$V_y = 0 \quad (\text{Constrained along the Y-axis})$$

Simultaneously, the Teensy 4.1 ingests the flawlessly smoothed, 1-degree resolution ball trajectory (θ_{ball}) from the Top-5 IR algorithm. The PID control loop dictates the lateral velocity (V_x) to directly mirror the incoming ball’s trajectory. This allows the robot to execute high-speed, closed-loop holonomic strafing parallel to the goal line:

$$V_x \propto \theta_{ball} \times K_{p_def}$$

By restricting movement to a 1D lateral plane and maintaining a locked compass heading facing the opponent’s field ($\omega = 0$), the robot functions as a highly reactive, impenetrable moving shield, mathematically optimizing its probability of intercepting high-speed incoming shots without ever breaching the field boundaries.

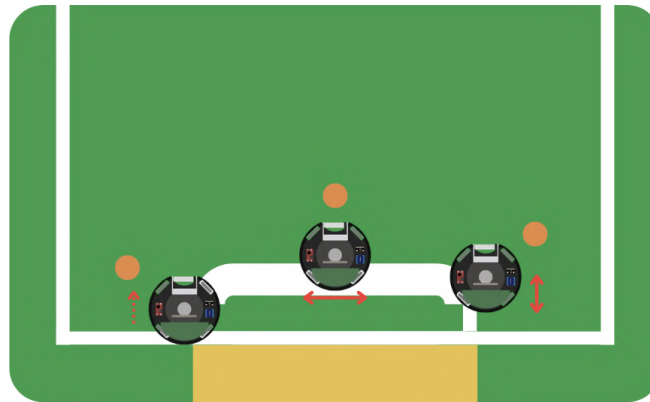


Fig. 40. The Dynamic Defensive State Machine. Active Vector Containment locks the robot’s longitudinal movement ($V_y = 0$) within the penalty zone, while the robot executes holonomic strafing (V_x) parallel to the goal line to mirror and intercept the ball’s trajectory.